ANNEX I: SAMOA—MEDIUM-TERM DEBT SUSTAINABILITY ANALYSIS[1]

The baseline and scenario analysis suggest that Samoa is at low risk of debt distress.

A. Baseline Scenario

Assumptions: The baseline scenario, described in Table 4, has medium-term growth expected at around 3½ percent a year, equal to that seen over the past decade. The continued growth reflects an anticipated step-up in tourism and foreign investment activities as recent economic reforms filter through to the rest of the economy. Despite impressive growth in recent years, foreign investment in Samoa has historically been well below that elsewhere in the Pacific, and extremely low compared with foreign investment in island economies elsewhere. Further structural reforms, including to increase the use of customary land, are expected to lead to increased foreign investment. With spending restraint generating a broadly balanced fiscal position, and supporting both domestic and external positions, the current account is expected to improve to around 1½ percent of GDP by 2010/11, and foreign reserves are expected to gradually rise to around 5 months of imports.

Debt Dynamics:[2] *Under this baseline, public debt and debt service indicators are all well below the levels typically associated with stress for countries with strong institutions and governance.*[3] Nominal public debt is expected to rise slightly to 39½ percent of GDP in FY2007/08 reflecting planned infrastructure spending, and then gradually fall to 34 percent of GDP by FY2010/11. With all public debt concessional, the NPV of external debt was less than 30 percent of GDP at end FY2005/06, and is expected to decline monotonically to around 22 percent of GDP by FY2010/11.[4]

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[1] Prepared jointly with World Bank staff.

[2] Only an external DSA has been prepared for Samoa. Domestic debt issuance has been minimal, with the authorities reporting domestic bond issues of around SAT 2 million (less than 0.2 percent of GDP). Given the highly shallow nature of Samoa's bond markets, the extent of domestic borrowing will likely remain very low for some time.

[3] Thresholds associated with international evidence of historical debt distress episodes in countries with strong institutions and governance are for the NPV of debt to: GDP above 50 percent; exports above 200 percent; revenue above 300 percent. For debt service, indicative ratios are for debt service to: exports above 25 percent; and revenue above 35 percent.

[4] Note that reflecting the importance of capital transfers, as well as the deficiency in Samoa's balance of payments data, there has been very large historical "residual" debt accumulation. Projected residual financing is smaller than historical patterns.

B. Sensitivity Analysis

Scenarios and bound tests do not suggest significant risks to Samoa's debt sustainability. Most scenarios and stress tests result in, at most, a temporary rise in the debt burden.[5] Under all standard stress (bound) test, the debt burden is below its FY2006/07 level within a decade. A permanent shock in which growth, inflation, the current account, and net FDI inflows permanently equal their average values over the past eight years (a "historical" scenario), would see a rising debt burden over time. This entirely reflects the high average non-interest current account deficit (3.5 percent of GDP), and low average FDI inflows (0.2 percent of GDP) over the past eight years. Indeed, endogenous debt dynamic under the historical scenario tend to reduce debt even faster than under the baseline scenario.[6] The importance of the large assumed current account deficit is demonstrated by an adjusted historical scenario where the current account and net FDI inflows are assumed to lie between the historical average and their baseline values. In this adjusted scenario, the debt burden declines generally over the horizon (Figure I-1). However, even in the unadjusted historical scenario the indicators of debt burden remain below the levels typically associated with debt distress in countries with strong institutions and governance.

Nonetheless, Samoa remains vulnerable to natural disasters. A scenario assuming a "cyclone" every five years starting from 2007/08,[7] which reduces GDP growth to zero and export growth to 60 percent of its baseline value, results in a relatively flat path for the NPV of debt indicators (with the NPV of debt rising very slightly relative to exports and falling very slightly relative to GDP). Debt service relative to exports rises over time. However, all indicators remain below the level associated with debt distress.

The analysis suggest that Samoa is at low risk of debt distress. This follows from standard debt indicators remaining well below the thresholds typically associated with potential distress, even under standard stress tests, throughout the period until FY2026/27.

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[5] Additional financing required under any of these scenarios or stress tests is assumed to be on concessional terms.

[6] Endogenous debt dynamics are those driven mechanically by movements in GDP growth, interest rates, inflation and the exchange rate.

[7] Samoa had three major cyclones between 1990 and 2005.

Table I-1. Samoa: External Debt Sustainability Framework, Baseline Scenario, 2006/07-2026/27 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2003/04	2004/05	2005/06			2006/07	2007/08	2008/09	2009/10	2010/11	2011/12	2006-11 Average	2016/17	2026/27	2012-26 Average
External debt (nominal) 1/	**47.8**	**39.3**	**39.4**			**37.9**	**39.4**	**39.0**	**37.0**	**34.0**	**30.9**		**16.2**	**2.5**	
o/w public and publicly guaranteed (PPG)	47.8	39.3	39.4			37.9	39.4	39.0	37.0	34.0	30.9		16.2	2.5	
Change in external debt	-2.8	-8.5	0.1			-1.5	1.5	-0.4	-2.0	-3.0	-3.1		-2.5	-0.7	
Identified net debt-creating flows	-8.2	-1.8	5.2			3.8	4.3	3.5	-0.8	-3.9	-3.6		-2.8	-2.1	
Non-interest current account deficit	**1.6**	**2.7**	**3.8**	**3.5**	**2.4**	**7.3**	**7.6**	**6.5**	**1.5**	**-1.7**	**-1.4**		**-1.1**	**-0.6**	-0.9
Deficit in balance of goods and services	21.4	24.4	28.9			32.1	32.6	31.6	26.9	23.4	22.7		19.6	17.7	
Exports	28.1	28.3	31.3			30.1	32.7	31.2	30.9	30.6	30.4		28.9	22.9	
Imports	49.5	52.7	60.2			62.1	65.3	62.8	57.8	54.0	53.0		48.5	40.7	
Net current transfers (negative = inflow)	-21.8	-25.3	-24.0	-21.1	2.7	-24.0	-24.8	-24.8	-25.2	-24.8	-24.5		-23.0	-20.6	-22.3
Other current account flows (negative = net inflow)	1.9	3.7	-1.2			-0.8	-0.3	-0.3	-0.2	-0.2	0.4		2.4	2.2	
Net FDI (negative = inflow)	**-0.7**	**0.5**	**0.9**	**0.2**	**0.5**	**-3.2**	**-2.3**	**-2.2**	**-1.3**	**-1.2**	**-1.2**		**-1.3**	**-1.4**	-1.3
Endogenous debt dynamics 2/	**-9.0**	**-5.0**	**0.5**			**-0.3**	**-1.0**	**-0.9**	**-1.0**	**-1.0**	**-0.9**		**-0.5**	**-0.1**	
Contribution from nominal interest rate	0.0	1.9	2.3			0.8	0.3	0.3	0.2	0.2	0.2		0.1	0.0	
Contribution from real GDP growth	-1.0	-2.5	-0.7			-1.1	-1.2	-1.1	-1.3	-1.2	-1.1		-0.6	-0.1	
Contribution from price and exchange rate changes	-8.1	-4.4	-1.1			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**5.4**	**-6.7**	**-5.1**			**-4.1**	**-1.7**	**-2.7**	**0.0**	**2.0**	**1.5**		**0.3**	**1.4**	
o/w exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
NPV of external debt 4/	...	...	28.9			26.8	26.4	25.8	24.3	22.4	20.6		11.8	2.2	
In percent of exports	...	...	92.5			89.3	80.8	82.8	78.6	73.2	67.8		41.0	9.4	
NPV of PPG external debt	**...**	**...**	**28.9**			**26.8**	**26.4**	**25.8**	**24.3**	**22.4**	**20.6**		**11.8**	**2.2**	
In percent of exports	...	...	92.5			89.3	80.8	82.8	78.6	73.2	67.8		41.0	9.4	
In percent of revenue	...	...	109.4			89.5	87.8	85.4	80.0	73.0	66.5		38.3	7.0	
Debt service-to-exports ratio (in percent)	...	10.5	11.1			6.5	4.1	4.2	4.6	4.9	4.7		3.7	1.7	
PPG debt service-to-exports ratio (in percent)	...	10.5	11.1			6.5	4.1	4.2	4.6	4.9	4.7		3.7	1.7	
PPG debt service-to-revenue ratio (in percent)	...	11.6	13.1			6.5	4.4	4.3	4.7	4.9	4.6		3.4	1.2	
Total gross financing need (billions of U.S. dollars)	2.9	25.1	34.9			27.4	31.9	29.0	8.9	-8.1	-7.7		-11.0	-32.2	
Non-interest current account deficit that stabilizes debt ratio	4.4	11.3	3.7			8.7	6.1	6.9	3.5	1.3	1.7		1.5	0.1	
Key macroeconomic assumptions															
Real GDP growth (in percent)	2.4	6.0	1.8	4.4	2.4	3.0	3.5	3.0	3.5	3.5	3.5	3.3	3.5	3.5	3.5
GDP deflator in US dollar terms (change in percent)	18.9	10.2	3.0	5.5	8.8	3.2	3.0	3.0	3.0	3.0	3.0	3.0	4.9	4.9	4.4
Effective interest rate (percent) 5/	0.0	4.5	6.2	1.8	2.8	2.3	0.8	0.7	0.6	0.6	0.6	0.9	0.6	0.7	0.6
Growth of exports of G&S (US dollar terms, in percent)	15.3	17.7	15.8	8.8	8.0	2.2	15.9	1.3	5.7	5.4	5.8	6.0	6.0	6.3	6.0
Growth of imports of G&S (US dollar terms, in percent)	14.3	24.3	19.8	12.3	7.4	9.7	12.1	1.9	-1.8	-0.5	4.7	4.4	6.7	6.7	6.1
Grant element of new public sector borrowing (in percent)	...	...	...	…	...	25.8	25.8	25.8	25.8	25.8	25.8	25.8	25.8	25.8	25.8
Memorandym item:															
Nominal GDP (millions of US dollars)	348.4	407.1	427.0			453.9	483.9	513.4	547.3	583.5	621.8		868.4	1976.8	

Sources: Samoan authorities and Fund staff estimates.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments devided by previous period debt stock.
6/ Historical averages and standard deviations are derived over the past 8 years.

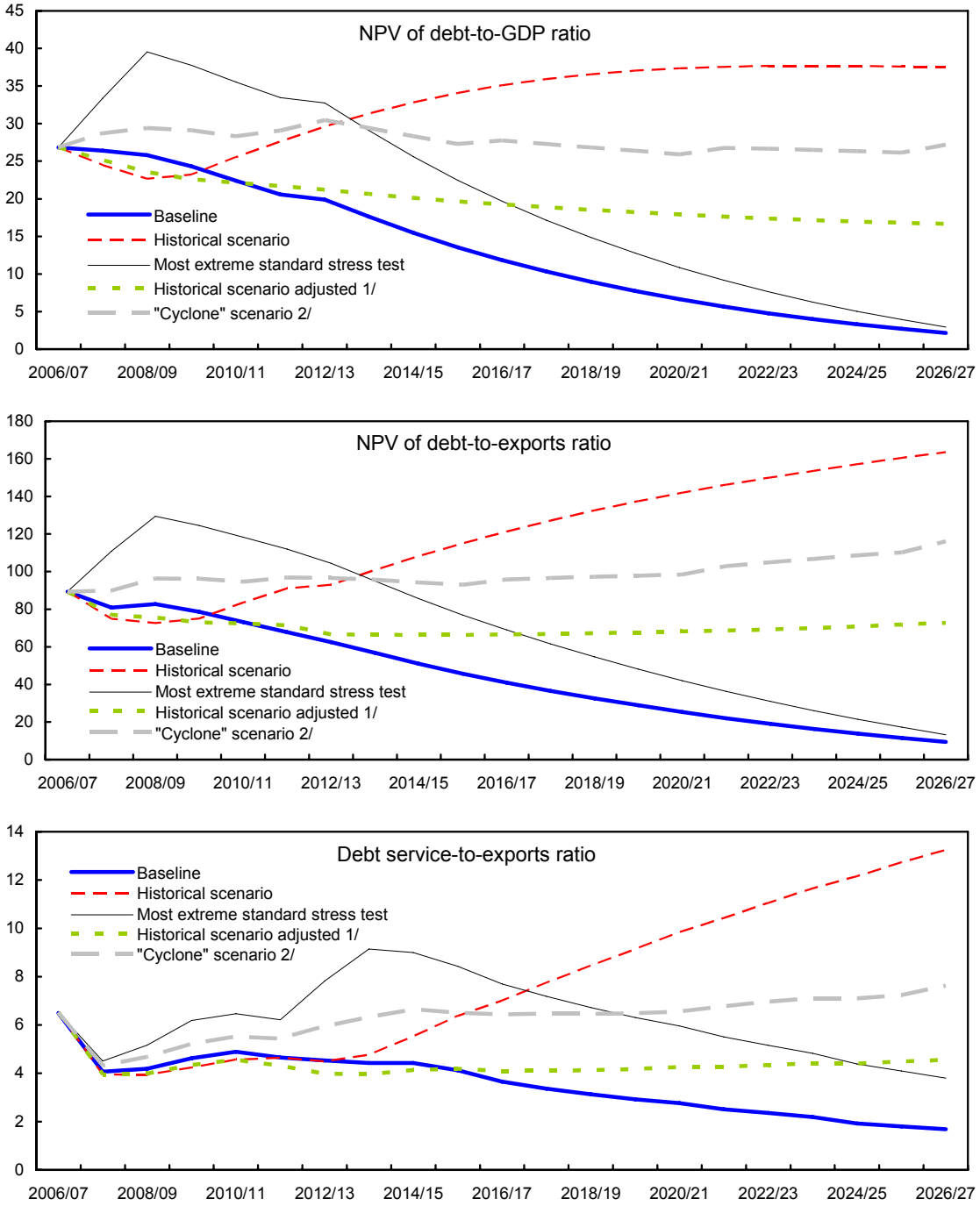
Table I-2. Samoa: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2006/07-26/27
(In percent)

		Projections						
	2006/07	2007/08	2008/09	2009/10	2010/11	2011/12	**2016/17**	2026/27
NPV of debt-to-GDP ratio								
Baseline	27	26	26	24	22	21	**12**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	27	24	23	23	26	28	**35**	38
A2. New public sector loans on less favorable terms in 2007-26 2/	27	27	26	25	23	21	**12**	3
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	27	27	26	25	23	21	**12**	2
B2. Export value growth at historical average minus one standard deviation in 2007-08 3/	27	30	32	30	28	26	**15**	3
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	27	28	29	28	25	23	**13**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 4	27	33	40	38	36	33	**20**	3
B5. Combination of B1-B4 using one-half standard deviation shocks	27	33	39	38	35	33	**19**	3
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/	27	37	36	34	32	29	**17**	3
B7. "Cyclone" scenario 6/	27	29	29	29	28	29	**28**	27
NPV of debt-to-exports ratio								
Baseline	89	81	83	79	73	68	**41**	9
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	89	75	73	75	84	91	**121**	164
A2. New public sector loans on less favorable terms in 2007-26 2/	89	82	84	81	75	70	**43**	11
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	89	81	83	79	73	68	**41**	9
B2. Export value growth at historical average minus one standard deviation in 2007-08 3/	89	104	118	113	107	100	**61**	13
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	89	81	83	79	73	68	**41**	9
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 4	89	102	127	122	116	110	**68**	13
B5. Combination of B1-B4 using one-half standard deviation shocks	89	111	129	125	118	112	**69**	13
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/	89	81	83	79	73	68	**41**	9
B7. "Cyclone" scenario 6/	89	90	96	96	95	97	**96**	116
Debt service ratio								
Baseline	6	4	4	5	5	5	**4**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	7	4	4	4	5	5	**7**	13
A2. New public sector loans on less favorable terms in 2007-26 2/	7	4	4	5	5	5	**4**	2
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	7	4	4	5	5	5	**4**	2
B2. Export value growth at historical average minus one standard deviation in 2007-08 3/	7	5	5	6	6	6	**7**	3
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	7	4	4	5	5	5	**4**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 4	7	4	5	6	6	6	**8**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	7	5	5	6	6	6	**8**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/	7	4	4	5	5	5	**4**	2
B7. "Cyclone" scenario 6/	7	4	5	5	6	5	**6**	8
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 7/	26	26	26	26	26	26	**26**	26

Sources: Samoan authorities and Fund staff estimates.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline, while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Baseline with a five-yearly "cyclone" that reduces GDP growth to zero and export growth by 40 percent.
7/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure I-1. Samoa: Indicators of Public and Publicly Guaranteed External Debt
Under Alternative Scenarios, 2006/07-2026/27
(In percent)



Source: Samoan Authorities and staff estimates.
1/ Historical scenario, except the current account and FDI flows are assumed to be an average of the past eight years average (historical scenario), and the baseline values.
2/ Baseline with a five-yearly "cyclone" that reduces GDP growth to zero and export growth by 40 percent.